Q1 2021 (31 Aug Shift 1)

Which of the following is *not* correct for relation R on the set of real numbers ?

(1) $(x, y) \in R \Leftrightarrow 0 < |x| - |y| \leq 1$ is neither transitive nor symmetric.

(2) $(x, y) \in R \Leftrightarrow 0 < |x - y| \leq 1$ is symmetric and transitive.

(3) $(x, y) \in R \Leftrightarrow |x| - |y| \leq 1$ is reflexive but not symmetric.

(4) $(x, y) \in R \Leftrightarrow |x - y| \leq 1$ is reflexive and symmetric.

Q2 2021 (27 Aug Shift 2)

Let $\mathbb{Z}$ be the set of all integers,

$A = \left\{(x, y) \in \mathbb{Z} \times \mathbb{Z} : (x-2)^2 + y^2 \leq 4\right\}$

$B = \left\{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x^2 + y^2 \leq 4\right\}$ and

$C = \left\{(x, y) \in \mathbb{Z} \times \mathbb{Z} : (x-2)^2 + (y-2)^2 \leq 4\right\}$

If the total number of relation from $A \cap B$ to $A \cap C$ is 2^p , then the value of p is :

(1) 16

(2) 25

(3) 49

(4) 9

Q3 2021 (27 Aug Shift 1)

If $A = \{x \in \mathbf{R} : |x - 2| > 1\}, B = \left\{x \in \mathbf{R} : \sqrt{x^2 - 3} > 1\right\},$

$C = \{x \in \mathbf{R} : |x - 4| \geq 2\}$ and $\mathbf{Z}$ is the set of all integers, then the number of subsets of the set $(A \cap B \cap C)^C \cap \mathbf{Z}$ is .

Q4 2021 (26 Aug Shift 1)

Out of all the patients in a hospital 89% are found to be suffering from heart ailment and 98% are suffering from lungs infection. If K% of them are suffering from both ailments, then K can not belong to the set:

(1) $\{80, 83, 86, 89\}$

(2) $\{84, 86, 88, 90\}$

(3) $\{79, 81, 83, 85\}$

(4) $\{84, 87, 90, 93\}$

Answer Key

Q1 (2) **Q2** (2) **Q3** (256) **Q4** (3)

Q1 (2)

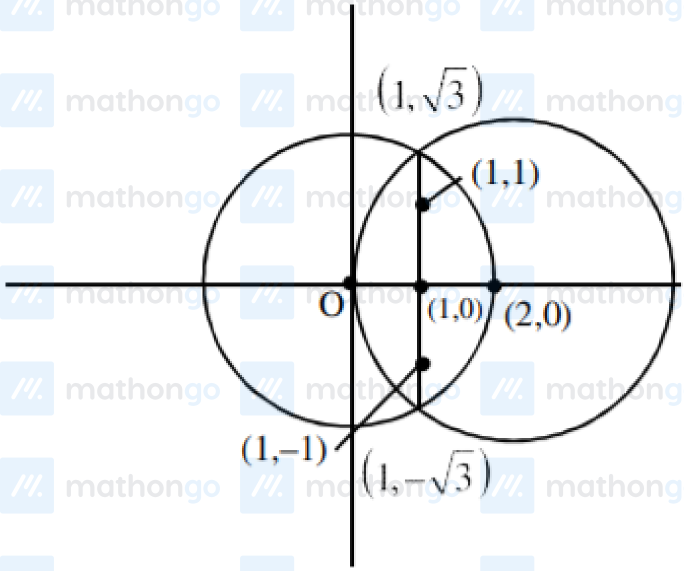
Note that $(1,2)$ and $(2,3)$ satisfy $0<|x-y| \leq 1$

but $(1,3)$ does not satisfy it so

$0 \leq |x-y| \leq 1$ is symmetric but not transitive

So, (2) is correct.

Q2 (2)



$(x-2)^2+y^2 \leq 4$

$x^2+y^2 \leq 4$

No. of points common in $C_1 \& C_2$ is 5.

$(0,0),(1,0),(2,0),(1,1),(1,-1)$

Similarly in C_2 \& C_3 is 5.

No. of relations $=2^{5\times5}=2^{25}$.

Q3 (256)

$\mathrm{A} = (-\infty, 1) \cup (3, \infty)$
$\mathrm{B} = (-\infty, -2) \cup (2, \infty)$
$\mathrm{C} = (-\infty, 2] \cup [6, \infty)$

So, $\mathrm{A} \cap \mathrm{B} \cap \mathrm{C} = (-\infty, -2) \cup [6, \infty)$

$\mathrm{z} \cap (\mathrm{A} \cap \mathrm{B} \cap \mathrm{C})' = \{-2, -1, 0, -1, 2, 3, 4, 5\}$

Hence no. of its subsets $= 2^8 = 256$.

Q4 (3)

$\mathrm{n}(\mathrm{A} \cup \mathrm{B}) \geq \mathrm{n}(\mathrm{A}) + \mathrm{n}(\mathrm{B}) - \mathrm{n}(\mathrm{A} \cap \mathrm{B})$

$100 \geq 89 + 98 - n(A \cup B)$

$\mathrm{n}(\mathrm{A} \cup \mathrm{B}) \geq 87$

$87 \leq \mathrm{n}(\mathrm{A} \cup \mathrm{B}) \leq 89$

Option (3)